

Instructions

The following .ppt is the template for the DRB and each team is **required** to use this format. Minor modifications made to the template are allowed, but keep in mind the score sheets are aligned with the template. It is also **required** to convert your .ppt to a .pdf (Maximum file size = 25MB), save it as “University_Name_2024_Baja_SAE_DRB” and submit to BajaSAE.net by the **date posted on BajaSAE.net** for the event you will attending **AND bring a printed copy to the design event.**

2024 Baja SAE Design Review Briefing (DRB)

School Name

Western Colorado University

2024 Car Number

42

OVERALL VEHICLE PRESENTATION

Top 2 – 8 Vehicle Goals

- Design and manufacture our school's first SAE Baja vehicle
- Finish manufacturing in time to attend the California competition in April 2024
- Abided by all the rules during manufacturing allowing us to pass Tech Inspection
- Supply the upcoming senior class with a comprehensive vehicle possible to use as a platform for next year

OVERALL VEHICLE PRESENTATION – GOALS AND REQUIREMENTS

Requirement	Design Target / Goal	Why this target was chosen	Result	Notes: (analysis, testing, additional details)
Curb Weight	650lbs	Decided upon value by previous team	???lbs	
Weight Bias Front / Rear	40/60	Prevent front diving through whoops and bumps	---	
Wheelbase	60"	Fits in the track	58"	
Overall Width (Outside edge of widest rims)	60"	Stable platform, narrow enough to fit through features	62"	
Overall Height (ground to top of roll cage)	70"	Enough height to fit in rules, result of ground clearance	72.125	
Ground Clearance	14"	Average of researched examples, better for rock crawling	15.5	
Tire Size	26"	Extra ground clearance for rocks	26"	These are the tires last years team order, so we designed suspension this year around them.
Front Suspension Type	Double Wish Bone A-arms	Simple implementation, freedom to control key variables	Double Wish Bone A-arms	
Front Suspension Travel	10"	Average travel distance from research	6"	Narrow toe box for increased freedom of suspension design
Rear Suspension Type	Trailing Arm	Simple yet sturdy design	Trailing Arm	
Rear Suspension Travel	11"	Keep balance with front, traction through bumps	--	
Steering Wheel Rotations lock to lock	1 Rotation	No need to remove hands from steering wheels	--	
Outside Turn Radius			--	
Top Speed	20 mph	Slow but will be raceable	--	
Acceleration Time (100ft, 150ft)	--		--	
Driveline Type (CVT, Sequential, eCVT)	CVT	Always in the power band	CVT	
4x4 Type (Chain, Shaft, Hydraulic...)	Shaft	Simplicity of front diff	Shaft	
Vehicle Life Target (miles/hours)	10 hours	Enough to test and get through competition ore time is a bonus	--	
Key Features				

OVERALL VEHICLE PRESENTATION

Chassis and Ergonomics High Level Summary

Chassis

- Goal: Design and manufacture a chassis capable of passing technical inspection to compete in dynamic events
- Researched similar designs for rear-braced SAE Baja vehicles
- Features include:
 - *A narrow toe box for increased suspension boundaries and performance*
 - *Custom bent and dimpled gussets for improved strength-to-weight ratio over conventional gusseting*
- Conducted FEA of chassis design for front and side impact
 - *Chassis capable of withstanding 40+mph front impact and 25mph side impact*

Ergonomics

- Goal: Create a driver seating position capable of seating both a 95th percentile male and 5th percentile female
- Researched basic body measurements for both males and females
- Features include:
 - *Kirkey aluminum race bucket seat with adjustable bolt spacing*
 - *Heel rests and pedal toe limiters for optimal driver foot and pedal position*
 - *Removable steering wheel for easy ingress and egress of vehicle*

OVERALL VEHICLE PRESENTATION

Drivetrain High Level Summary

Drivetrain

Goal

- *Design and manufacture a drivetrain capable of passing technical inspection to compete in dynamic events.*

Stock parts being used:

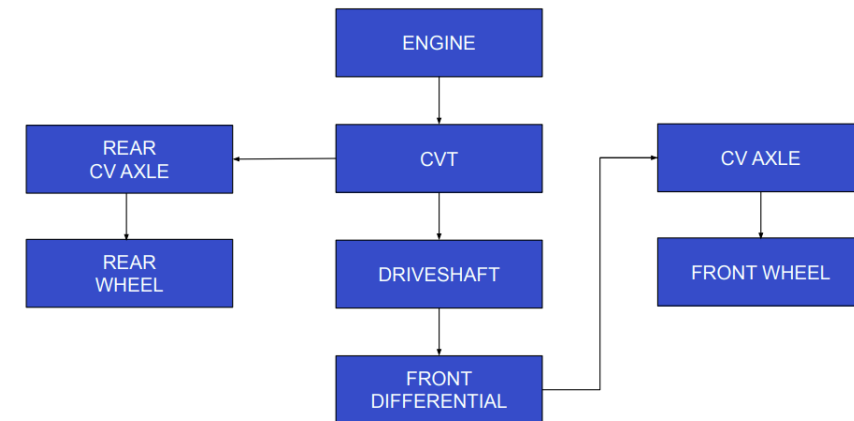
- *2001 Polaris Sportsman 500 Transmission*
- *2004 Polaris Sportsman 500 Front Differential*

Features Include:

- *Custom length driveshaft created from u-joints off a Polaris Sportsman*
- *Shielding was created to meet HROE requirements while being as compact as possible*
- *PVC finger guard around the prop shaft*
- *Expanded metal finger guard around the clutch*
- *Custom hoops around the prop shaft*
- *Floor panels were modified to allow for rigid mounting points to the chassis for hoops*

Analysis:

- *The driveshaft diameter was chosen based on the torque produced by our engine*
- *HROE guard materials were chosen based on rules as well as given a factor of safety*



Flow chart showcasing power transfer

OVERALL VEHICLE PRESENTATION

Suspension, Steering and Brakes High Level Summary

Suspension

Goal:

- *Design a suspension focusing on ride height, roll center, KPI, bump steer, camber and scrub radius*

Features Included:

- *Front: Double wishbone A-arm*
- *Rear: Trailing arm*

Analyses:

- *Racing Aspirations simulation*

Steering

Goal:

- *Implement a complete reverse Ackermann front steering system capable of enduring competition.*

Features include:

- *Stiletto C42-344 Rack N Pinion Vertical 15:1*
- *11 1/4 C-C 3/8 Rod Ends*

Analyses:

- *We choose brakes that have stopping power within our desired range*

Brakes

Goal:

- *Purchase and implement a braking system in both the front and back*

Features Include:

- *Wilwood master cylinders and proportioning valve, Polaris Sportsman calipers, flexible steel braided brake lines*

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FULL VEHICLE ENGINEERING

Benchmarking and Research

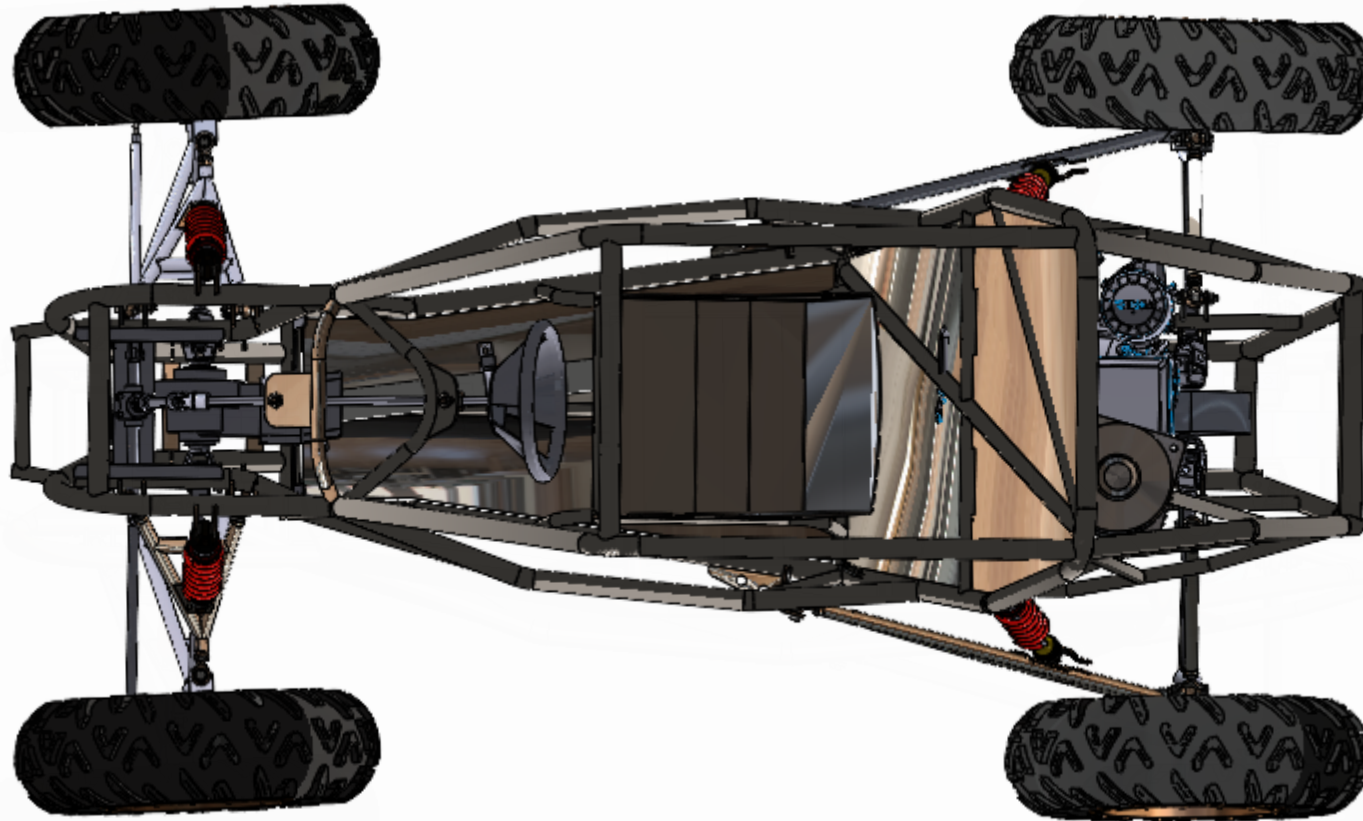
- **How did you achieve the measurable data for your vehicle goals?**
 - *Competitor vehicles*
 - *Other vehicles in the market today*
- **What evaluation/testing was completed?**
 - *N/A - this was our team's first year completing many of these tasks*
- **How do your vehicle goals compare to the benchmark data collected?**
 - *N/A*
- **What research was conducted to help identify potential options that could achieve the vehicle goals**
 - *Research was done on side by sides, ATVs, and other Baja vehicles to identify potential options for our vehicle*
- **List out all the options that were considered and what goal(s) it could potentially help achieve**
 - *Research was conducted on similar vehicles we could find on the market or from other universities, as this was our first year completing this project*

FULL VEHICLE ENGINEERING

Component & Subsystem Selection

- **How did you select the components/subsystems used? / What components/subsystems did you ultimately select for your vehicle?**
 - *The budget was the largest consideration when choosing components/subsystems*
 - *CV transmission, front differential, and driveshaft came from a donated ATV*
 - *Due to the heavier weight of our car, we purchased custom-tuned Elka Suspension STAGE 4 shock towers to meet our desired ride height and suspension performance*
 - *Wilwood disk brake parts were chosen for their all-around optimum stopping power*
 - *Space was the primary constraint for steering. We had to select a compact rack and pinion location to fit our limited toe box size. Everything else was designed around this space limitation to ensure proper integration with other subsystems.*

FULL VEHICLE ENGINEERING



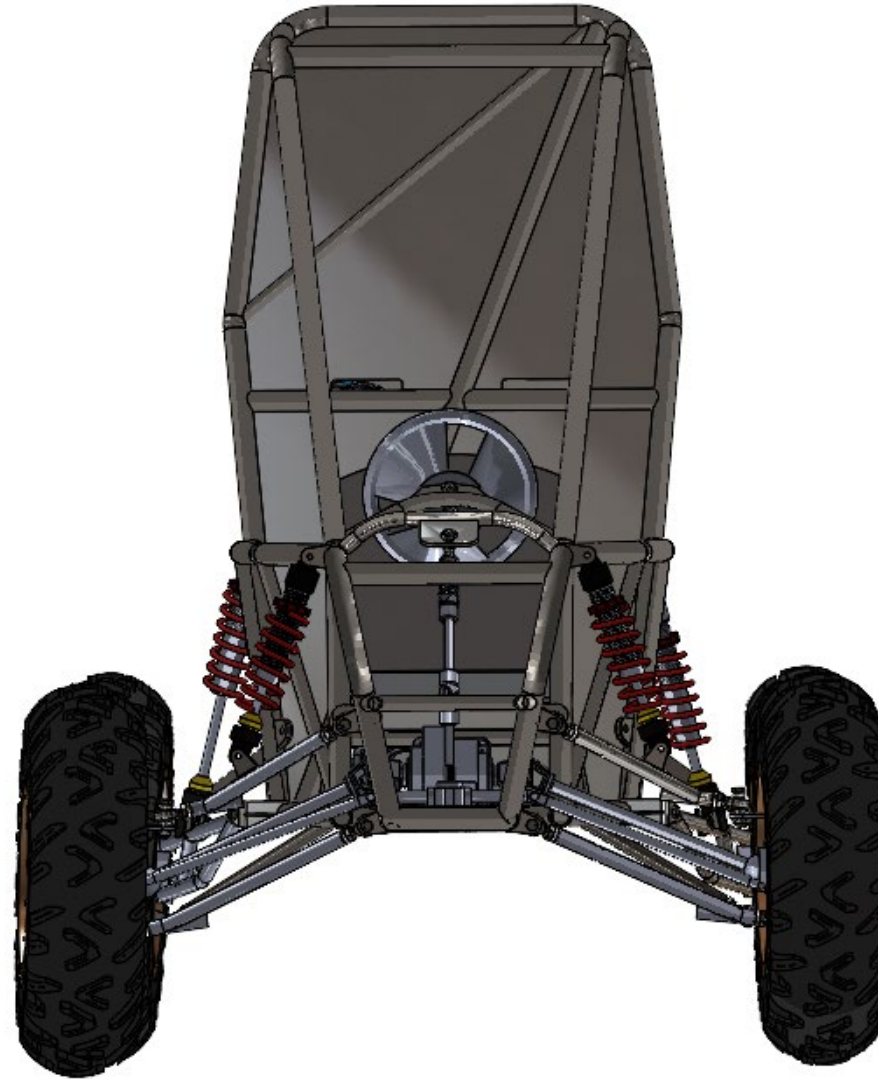
Top View

FULL VEHICLE ENGINEERING



Left Side View

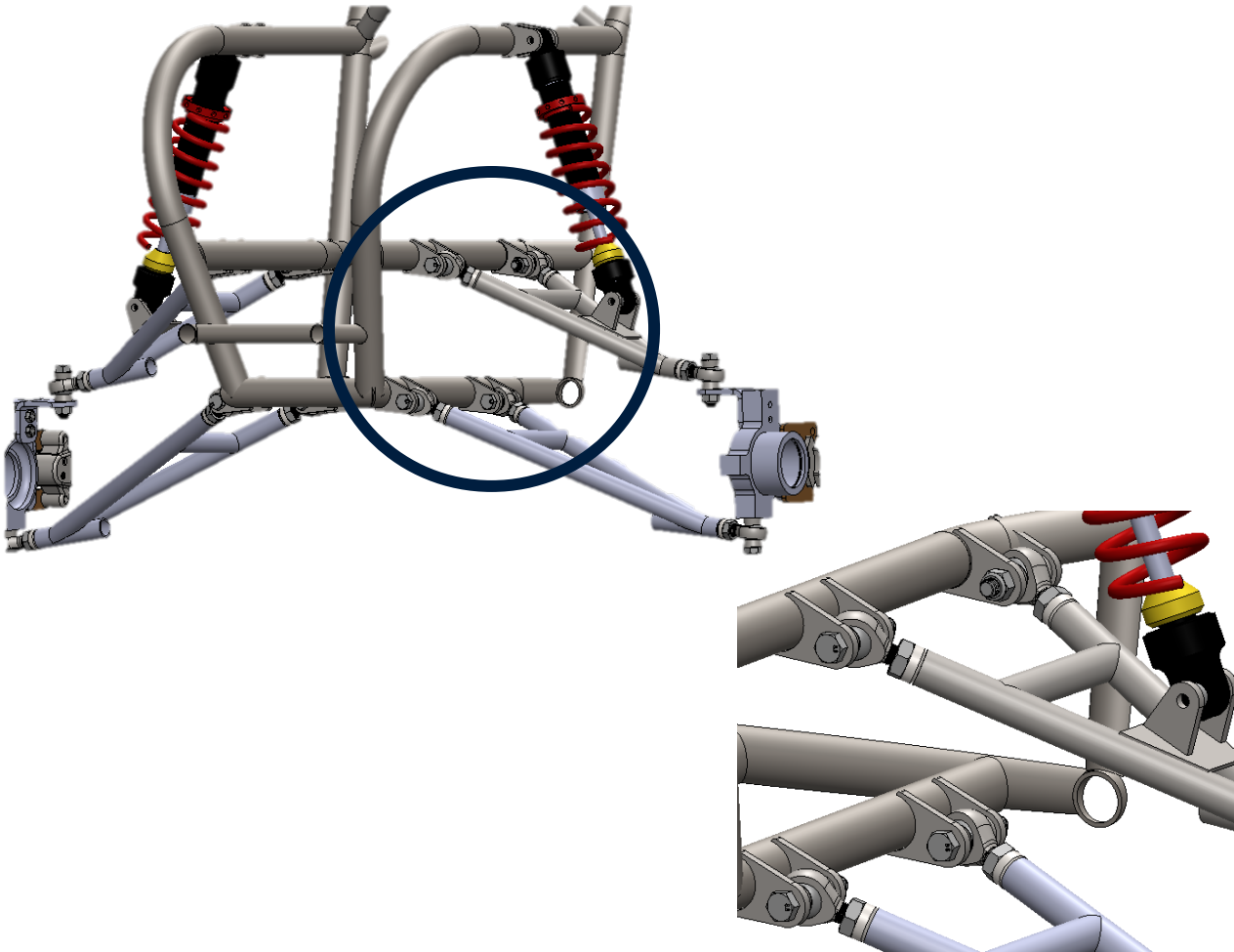
FULL VEHICLE ENGINEERING



Front View

FULL VEHICLE ENGINEERING

Suspension points → Chassis CAD



Suspension points → Chassis Explanation

Integration

- *1/4" plasma cut steel tabs*
- *Tig welded to the chassis*
- *Heim joints attached with grade 8 bolt and lock nut*

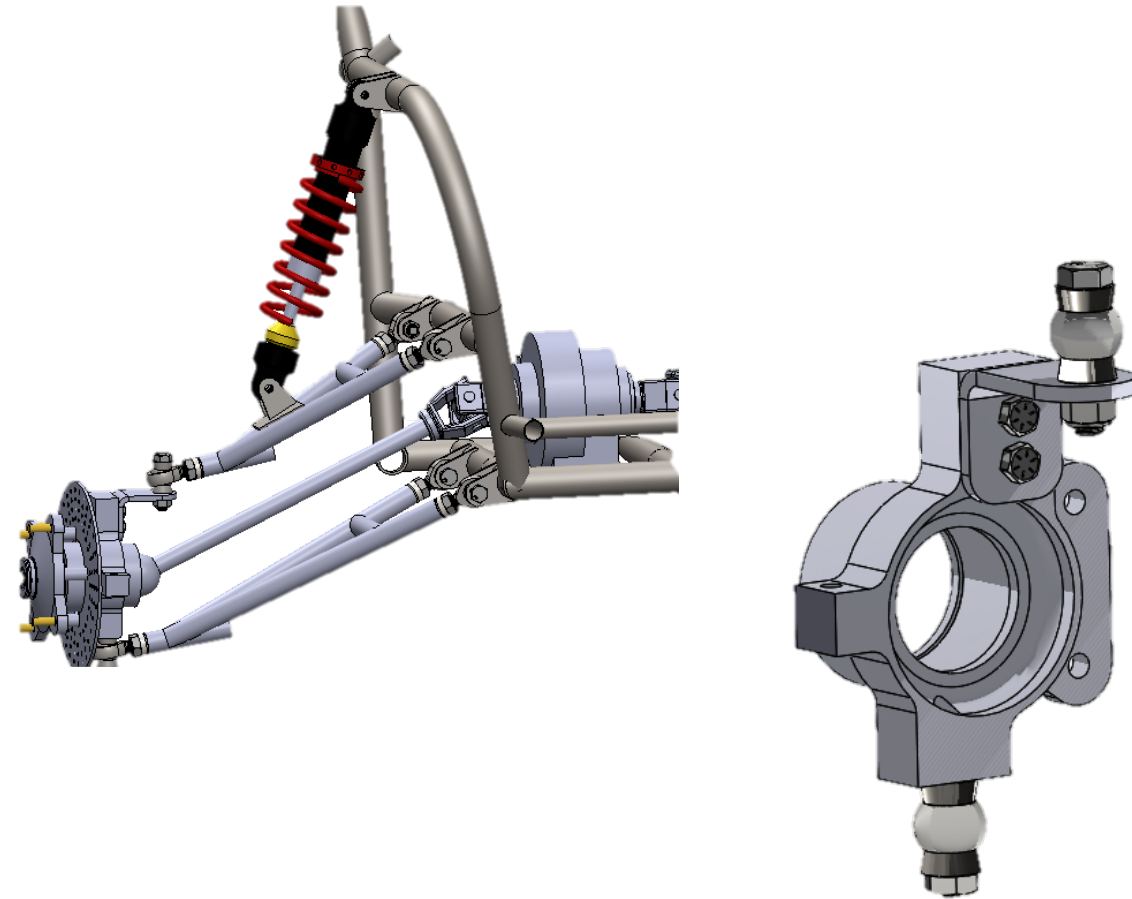
Explanation

- *Considerations arose with our previous tabs only being only 1/8" thick*
- *We increased the plate thickness to ensure the tabs could withstand the forces applied from dynamic events that the car will encounter as well as side and front impact forces*

Front Left View

FULL VEHICLE ENGINEERING

Front differential → Hubs CAD



Front differential → Hubs Explanation

Integration

- *Polaris Sportsman 500 2001 front differential*
- *Custom hub carriers were manufactured out of aluminum stock on the CNC machine*
- *The custom hub carrier houses a bearing and snap ring for hub attachment*
- *The custom hub carrier provides enough room for the CV axle to pass through and connect with the hubs*

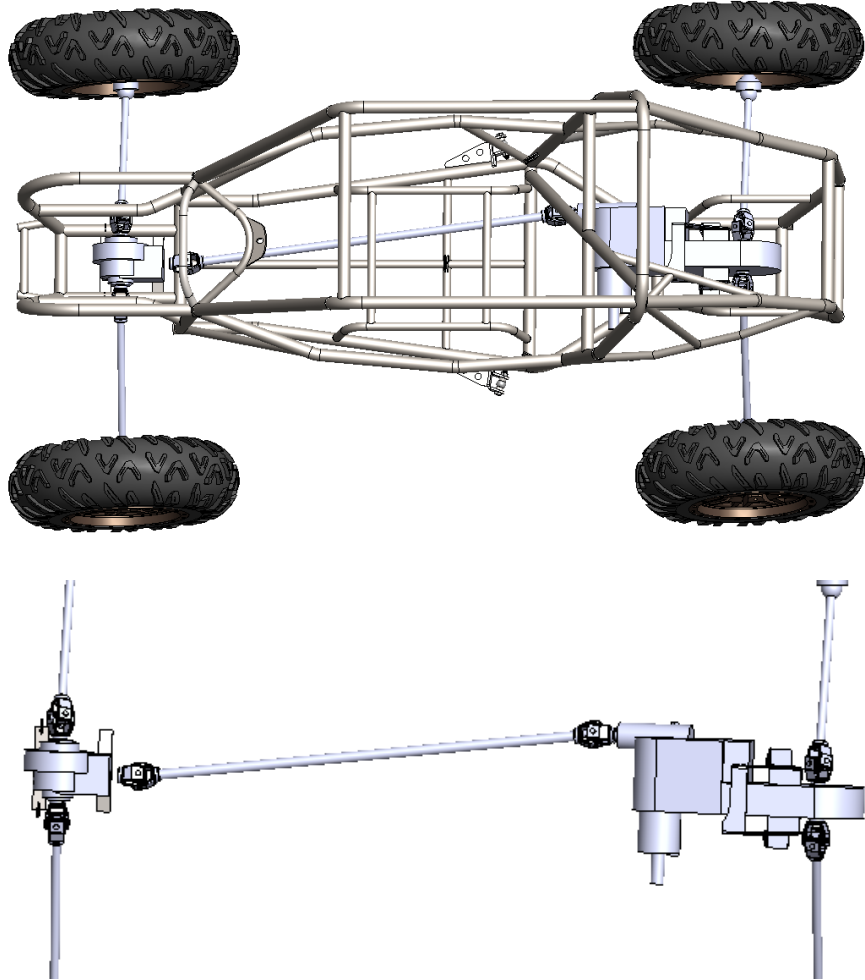
Explanation

- *To transmit power from the front differential to the hubs, we customized CV axles by cutting and sleeving them to match the length required by our suspension and hubs.*
- *As a team tackling this for the first time, we saw this approach as beneficial because it allowed us to become familiar with the parts and understand how they function.*

Front Right View

FULL VEHICLE ENGINEERING

Transmission → Driveshaft → Front Differential CAD



Transmission → Driveshaft → Front Differential Explanation

Integration

- *U-joints on each end of the drive shaft*
- *Custom length driveshaft*

Explanation

- *The U-joints we used were sourced from a used Polaris ATV*
- *The driveshaft was sourced from a used Polaris ATV and then cut, extended, and welded to our custom length*

Top View

FULL VEHICLE ENGINEERING

Full Vehicle / Subsystem Integration Testing

- **What physical subsystem or vehicle testing was conducted to demonstrate that your vehicle was capable of meeting your goals?**
 - *Suspension testing was our biggest goal for the entire car. The testing of suspension was what drove the entirety of our project.*
- **What measurable data was collected and how did that compare to the measurable goals you set?**
 - *The team would measure the static ride height and drive the car in rough conditions to determine how the car handles in real-world applications.*
- **How did the data that was collected correlate back to your analysis?**
 - *No data has been collected at this time.*
- **Demonstrate how the testing results impacted the design of the vehicle. Show test results before and after a design change and explain how the change improved meeting your design goals**
 - *No data or modifications have been made at this time.*

FULL VEHICLE ENGINEERING

Assessment Of Vehicle Goals

- **Compare the top vehicle goals to what the team actual achieved**
 - *The top vehicle goal was to finish designing and manufacturing the first vehicle at Western Colorado University. In addition to this, our goal is to make it to the competition. Throughout the year, serious progress has been made, and the opportunity to make it to competition is still present.*
- **What goals did the team succeed at and what goals did the team come up short**
 - *The team is on track to have the car in competition for the 2024 season.*
- **For the goals that you came up short on what was the root cause**
 - *If the team does not make it to the competition, it was due to a small team with a lack of experience that was not able to foresee issues that arose and fix the issues.*

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CHASSIS / ERGONOMICS

Requirements

- Meet all SAE rules
- Fit our largest and smallest team member as a driver
- Adjustable seat and safety harness, as the steering and pedal box will not be adjustable
- Space between the rear members and the space in the toe box must be large enough to encase our large transmission and front differential
- Headspace must be large enough to fit our tallest driver with a helmet and still conform to the regulations

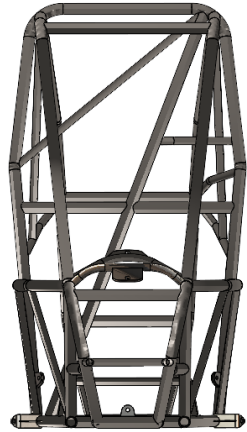
Research and Development

Most of our research and development for the frame was based on the requirements stated in the SAE Baja rulebook. Shared reports and papers from other universities were also used.

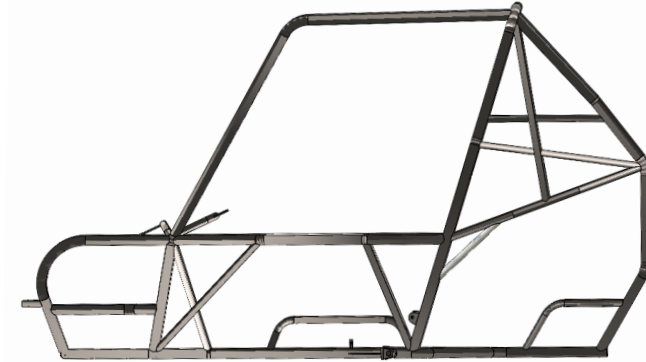
We met with our partnership school, CU Boulder, to gather advice for starting this project, and we also looked at the reports they shared online for design and manufacturing ideas.

CHASSIS / ERGONOMICS

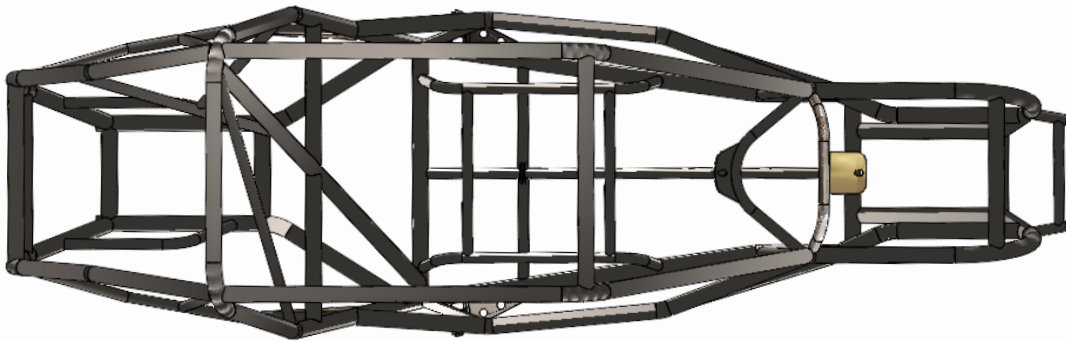
Chassis CAD – Front View



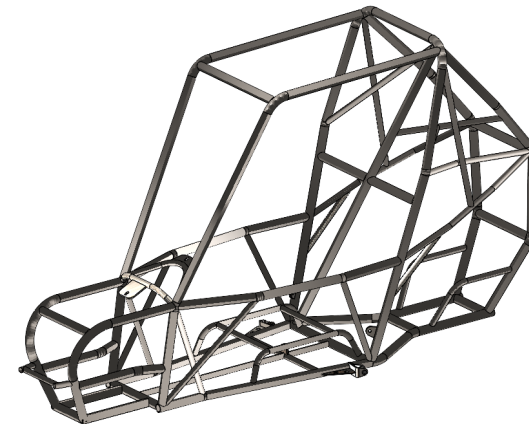
Chassis CAD – Left Side View



Chassis CAD – Top View

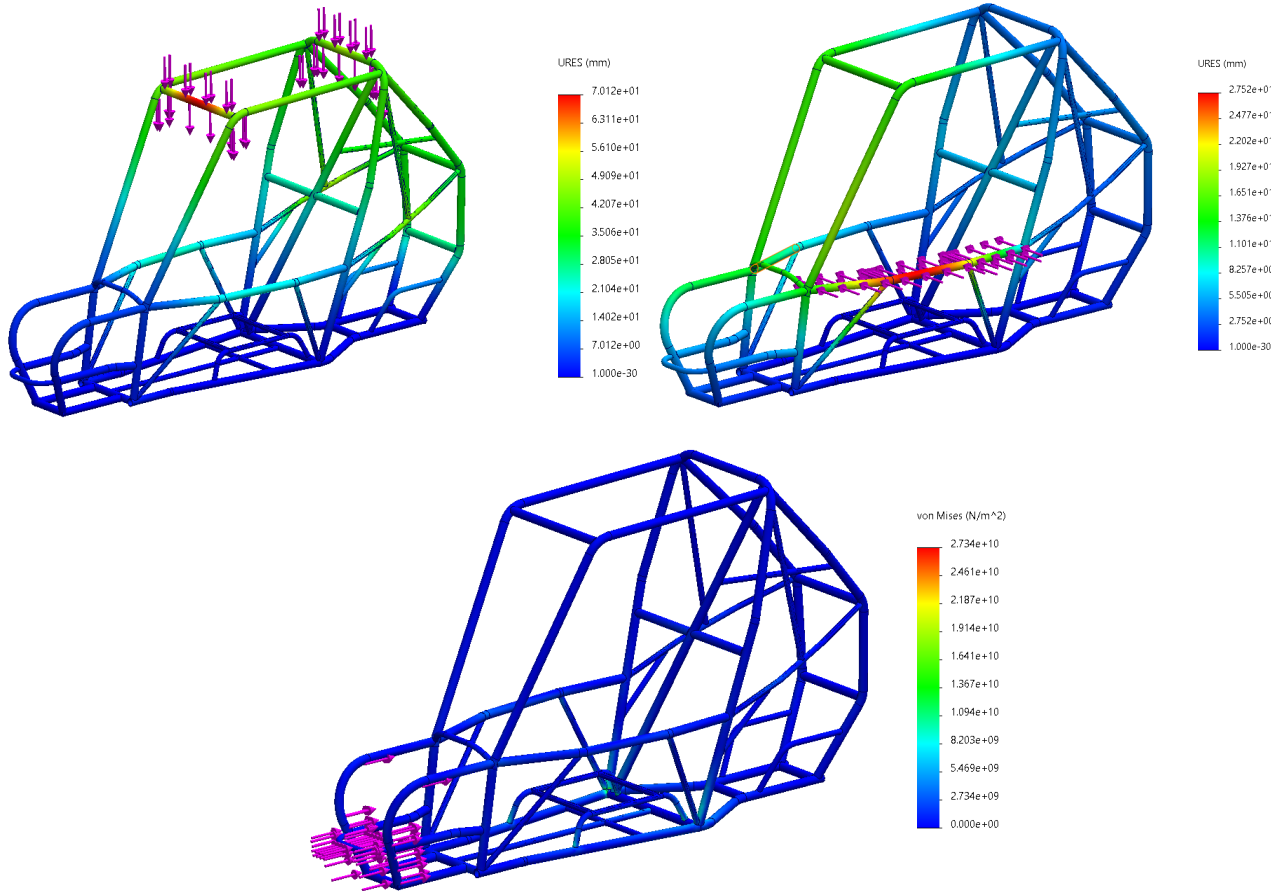


Chassis CAD – Isometric View



CHASSIS / ERGONOMICS

Chassis Data Collection for Analysis



Chassis Analysis Summary

- *Applied force for front FEA analysis was calculated using our 800lb estimation for our total vehicle weight, as well as a 40mph crash, assuming a max speed of our or other teams' cars*
- *Max deformation for a front hit was negligible*
- *Max deformation for the top impact was 7cm*
- *Max deformation for a 20mph side hit was 3cm*
- *All calculated von mises stresses were below the critical stress rating for our material*

CHASSIS / ERGONOMICS

Chassis Testing / Validation

No testing has been performed on the chassis at this time.

Chassis Correlation of Analysis and Testing

Weld Testing:

Our weld testing revealed that joint failures occurred when bent beyond 7cm from their original position. This aligns with our FEA simulations, which showed a maximum deformation of 7cm. This confirms that our vehicle can withstand impacts at speeds of 40mph or less without experiencing broken joints, ensuring both frame integrity and driver safety.

CHASSIS / ERGONOMICS

Chassis Manufacturability

The chassis was custom-manufactured by us in our on-campus workshop. Chassis features that made this possible include the following:

- *Tube dimensions that follow the rules while also being the correct size to fit in our tube bender without buckling*
- *Every tube end was coped to allow for precise fit-ups and stronger welds at joints*

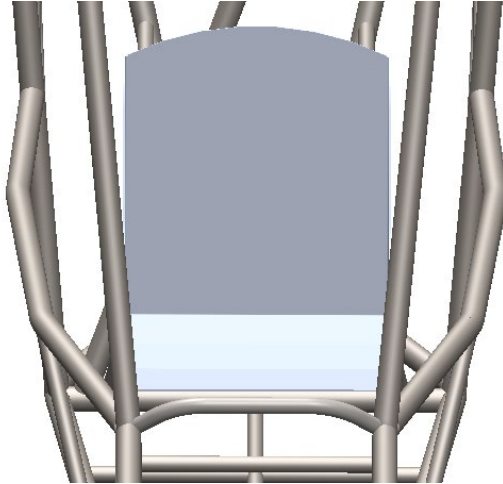
Chassis Serviceability

Features of our chassis that make it easy to service are:

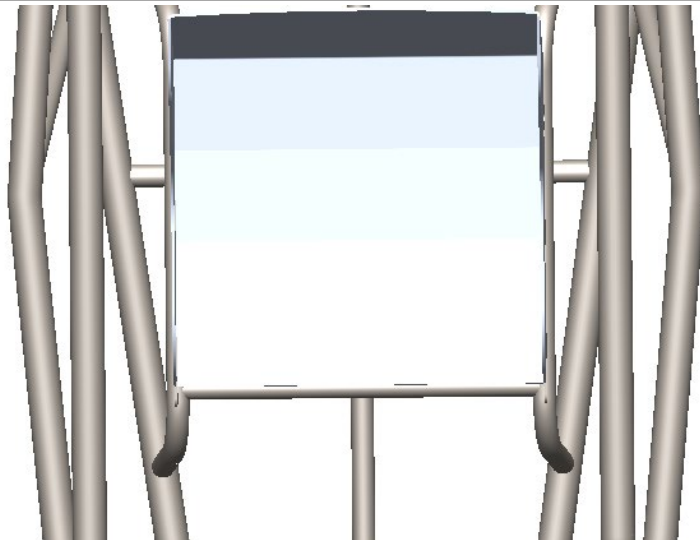
- *A removable member in the rear which makes it possible for us to easily access our engine and transmission*
- *Every component of our vehicle is removable as they are all attached via tabs welded to the frame, this allows us to remove any components interfering with our path to service the chassis*

CHASSIS / ERGONOMICS

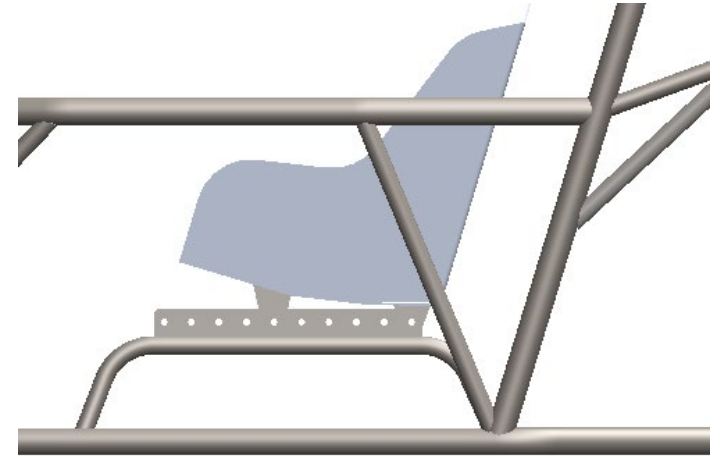
Ergonomics CAD – Front View



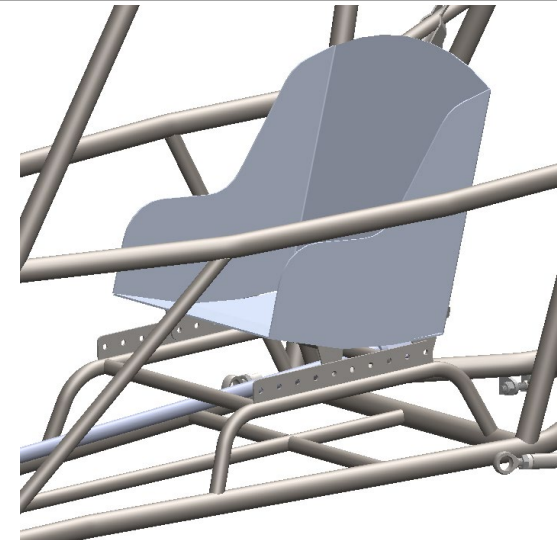
Ergonomics CAD – Top View



Ergonomics CAD – Left Side View



Ergonomics CAD – Isometric View



CHASSIS / ERGONOMICS

Ergonomics Data Collection for Analysis

Research:

According to the rules, our vehicle's ergonomics must be useable for all people within the range of a 5th percentile female and a 95th percentile male. According to our research, the following data was found pertaining to these requirements:

5th percentile female:

- 5' 0" tall
- 27" legs
- 33" torso

95th percentile male:

- 6' 2" tall
- 33.3" legs
- 40.7" torso

Ergonomics Analysis Summary

Determining distances:

Using measurements from our research, we gathered data to guide our CAD measurements. We assessed the distance from the back of the seat to the pedals in both the closest and furthest seat settings to ensure the driver's reachability to the pedals. Additionally, we measured from the bottom of the seat to the top members of our chassis to ensure clearance for even the tallest drivers.

CHASSIS / ERGONOMICS

Ergonomics Testing / Validation

Testing:

Using the heights derived from our research, encompassing the 5th percentile female and 95th percentile male, we enlisted team members or individuals from our class who matched these measurements to test the ergonomics of our vehicle. We adjusted the seat positioning for each person and ensured they could comfortably reach the pedals while maintaining adequate head clearance between their helmet and the nearest frame member.

Ergonomics Correlation of Analysis and Testing

Results:

Our testing with individuals sitting in the driver's seat revealed that our initial CAD measurements closely matched the requirements for comfortably accommodating a 5th-percentile female and 95th-percentile male. The variance between our expected measurements based on CAD and the actual results from our tests was negligible.

CHASSIS / ERGONOMICS

Ergonomics Manufacturability

Excluding the bucket seat, which was bought online from a racing company, all aspects of ergonomics were manufactured in-house by members of our team. The seat mount was made using a plasma table for the adjustable hole mounts, and the tube mount was welded together after being bent and notched.

Ergonomics Serviceability

The bucket seat is adjustable and removable making it very easy to service or replace. This also allows us to easily access components otherwise covered by the seat if we need to. The bucket seat has 5 different adjustability settings allowing for a wide range of driver sizes to fit comfortably in the seat. Our padded seat cover is also easily removable allowing for easy replacement or serviceability.

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DRIVETRAIN

Requirements

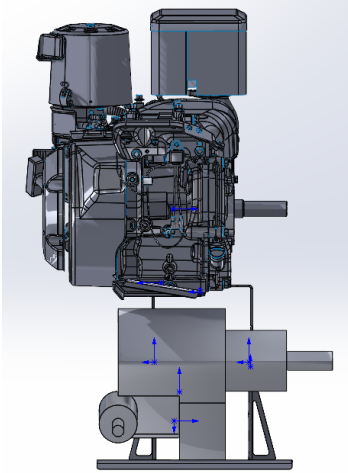
- Adhere to SAE Baja rules
- 4-wheel drive
- Utilize the transmission and front differential obtained from a pre-owned Polaris Sportsman 500 ATV to optimize cost efficiency within our budget.

Research and Development

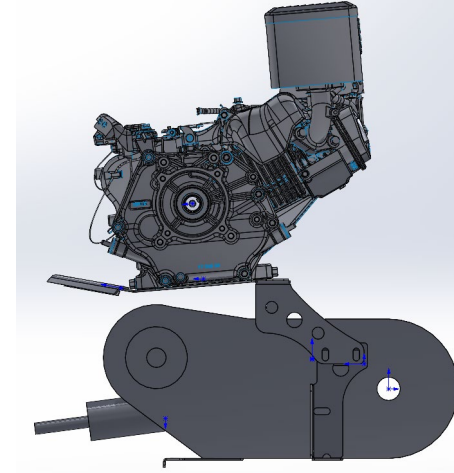
Similar to our previous research and development efforts, much of our drivetrain research and development stemmed from studying existing products on the market similar to our design, along with exploring documentation shared by other universities participating in Baja competitions.

DRIVETRAIN

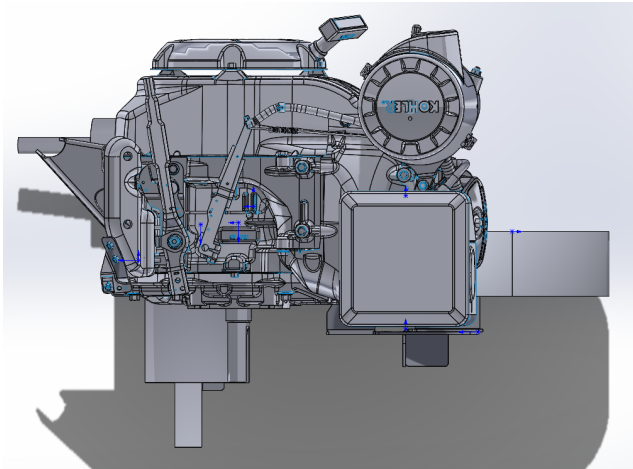
CVT / Transmission CAD – Front View



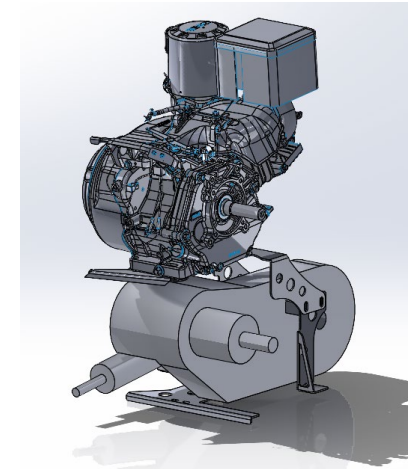
CVT / Transmission CAD – Left Side View



CVT / Transmission CAD – Top View



CVT / Transmission CAD – Isometric View



DRIVETRAIN

CVT / Transmission Collection for Analysis

No data was collected on our transmission after it was selected. Before selecting our CVT, we made sure to look at CVTs with the following requirements:

- A high and low gear
- Ability for 4-wheel drive via a front differential
- Ideally meant to be paired with the front differential we choose

CVT / Transmission Analysis Summary

Although no data was collected, it was chosen specifically because it includes all the features we were looking for and could integrate well with the given engine.

DRIVETRAIN

CVT / Transmission Testing / Validation

Since the CVT was sourced from a pre-owned ATV and not manufactured by us, our testing primarily focused on verifying its operational status. Once confirmed, our efforts shifted towards designing and constructing custom wiring and power-sourcing methods in-house.

CVT / Transmission Correlation of Analysis and Testing

After choosing a CVT that met our needs and verifying it was still working after being used in an ATV before us, we were able to install it and verify that our initial requirements were true to what our physical vehicle needed now that it was manufactured.

DRIVETRAIN

CVT / Transmission Manufacturability

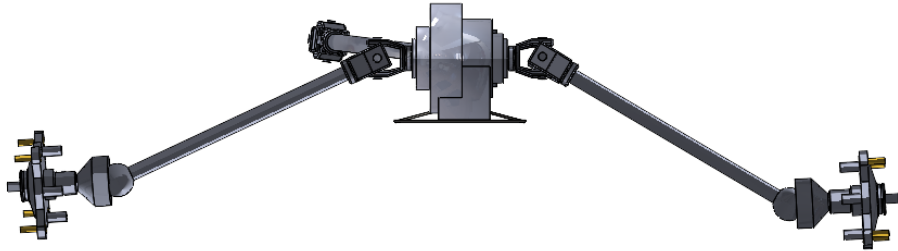
*Not applicable as our CVT transmission was sourced from a Polaris Sportsman ATV

CVT / Transmission Serviceability

Unless a basic external component of the CVT malfunctions, servicing it won't be straightforward due to its complexity and our lack of manufacturing it. However, both the CVT and engine are easily removable, facilitating replacement with a secondary CVT should the original one fail.

DRIVETRAIN

Front Drive, Prop Shaft, Front Hubs and Axles CAD – Front View



Front Drive, Prop Shaft, Front Hubs and Axles CAD – Left Side View



Front Drive, Prop Shaft, Front Hubs and Axles CAD – Top View

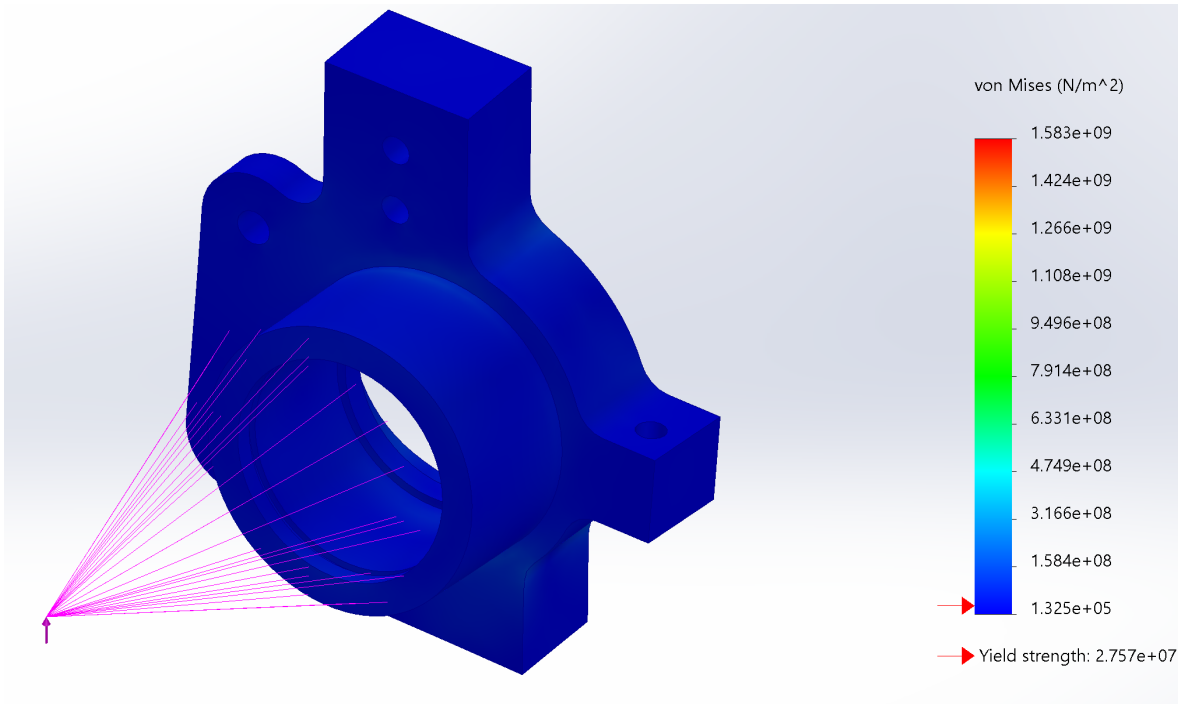


Front Drive, Prop Shaft, Front Hubs and Axles CAD – Isometric View



DRIVETRAIN

Front Drive, Prop Shaft, Front Hubs and Axles Data Collection for Analysis



Front Drive, Prop Shaft, Front Hubs and Axles Analysis Summary

- Front differential from a 2001 Polaris Sportsman 500
- Custom machined aluminum hub carriers
- Cut, sleeved, and extended CV axels
- Cut and extended prop shaft

DRIVETRAIN

Front Drive, Prop Shaft, Front Hubs and Axles Testing / Validation

No testing has been performed on the front drive shaft, hubs, or axles.

Front Drive, Prop Shaft, Front Hubs and Axles Correlation of Analysis and Testing

No testing has been performed on the front drive shaft, hubs, or axles.

DRIVETRAIN

Front Drive, Prop Shaft, Front Hubs and Axles Manufacturability

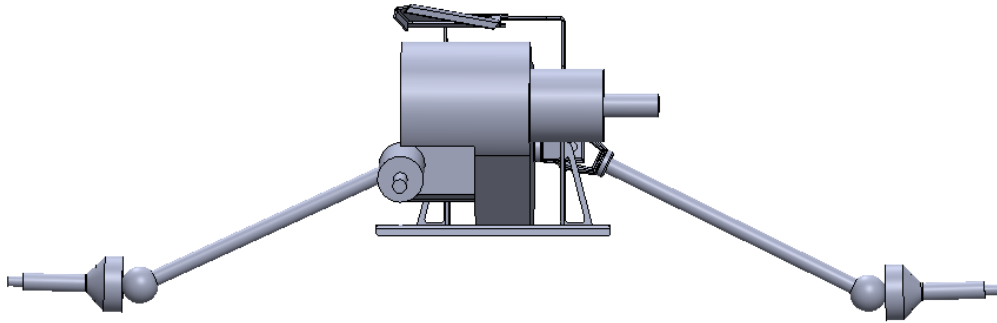
The prop shaft, front hubs, and axles were all fabricated in-house by our team members. Axles underwent cutting, sleeving, and welding to meet our specifications, while the prop shaft was extended to the required length. Additionally, our team utilized CAD, CAM, and a 3-axis CNC mill for designing and manufacturing the front hubs. The only component in this section of the drivetrain that we didn't manufacture was the front differential.

Front Drive, Prop Shaft, Front Hubs and Axles Serviceability

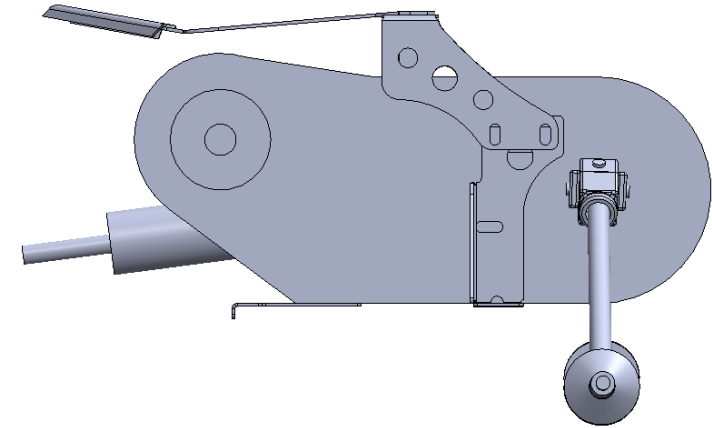
Servicing the front differential, front hubs, and axles poses challenges due to their intricate manufacturing process, requiring CNC mills or precise machinery. Consequently, they are not field-serviceable if they fail. In contrast, the prop shaft is simpler to service. However, all mentioned components are easily removable, facilitating access to other parts that may be obstructed by them.

DRIVETRAIN

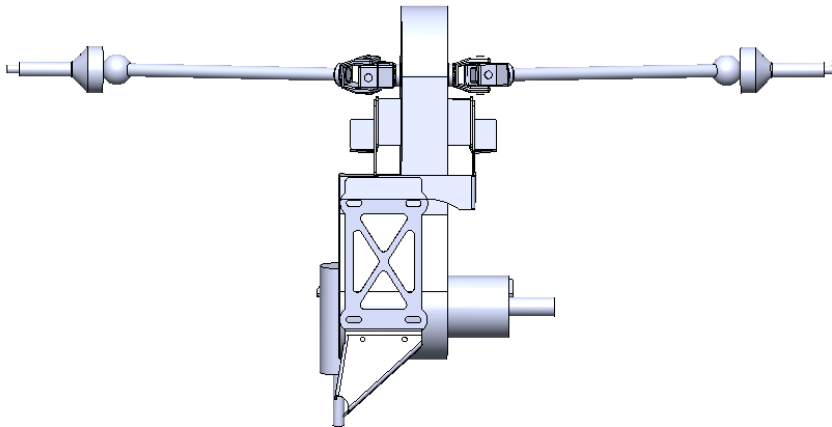
Reduction Box, Rear Hubs and Axles CAD – Front View



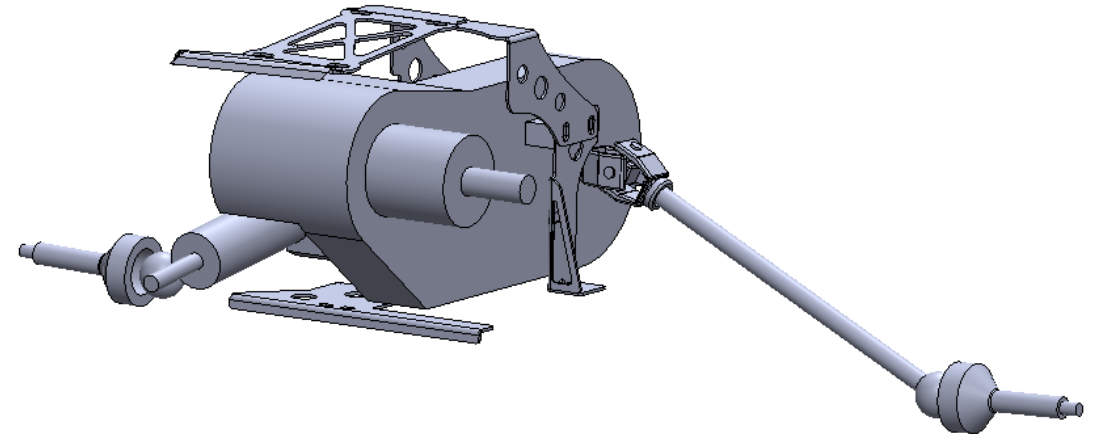
Reduction Box, Rear Hubs and Axles CAD – Left Side View



Reduction Box, Rear Hubs and Axles CAD – Top View



Reduction Box, Rear Hubs and Axles CAD – Isometric View



DRIVETRAIN

Reduction Box, Rear Hubs and Axles Data Collection for Analysis

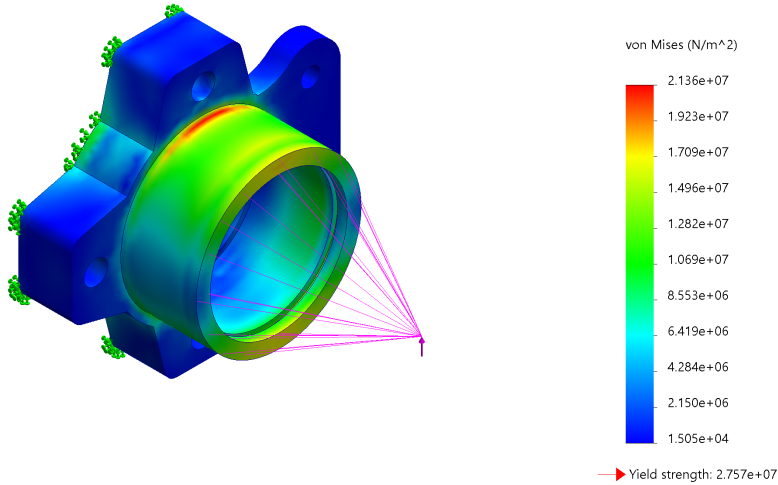
Testing has not been performed on the reduction box, rear hub, or axles.

Reduction Box, Rear Hubs and Axles Analysis Summary

The reduction box is integrated into our CVT, streamlining energy transfer and conserving rear space. Rear hubs, like the front ones, were machined in-house from aluminum. The rear axles were also customized internally to align with our requirements, mirroring the approach taken with the front axles.

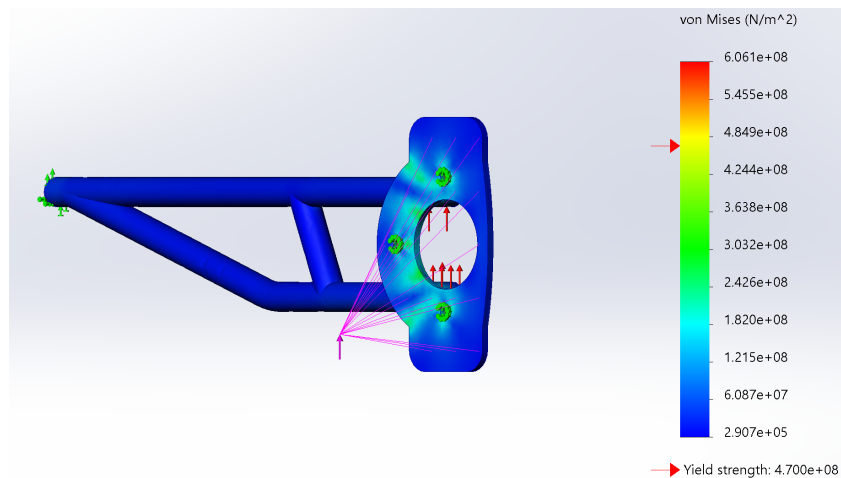
DRIVETRAIN

Reduction Box, Rear Hubs and Axles Testing / Validation



Reduction Box, Rear Hubs and Axles Correlation of Analysis and Testing

The rear hubs are able to withstand the forces expected.



DRIVETRAIN

Reduction Box, Rear Hubs and Axles Manufacturability

The rear hubs were custom-designed and manufactured in-house by members of our team using CAD, CAM, and the CNC mill. The rear axles are similar to the front axles; using the CV axles we ordered, we then cut, extended, sleeved, and welded the axles to our desired length. The only part of this section of the drivetrain that we did not custom-make ourselves was the reduction box, as it was integrated into the CVT we selected.

Reduction Box, Rear Hubs and Axles Serviceability

Serviceability for this section of the drivetrain is very similar to the previous, as the majority of these components are too complex to service in the field, without having a replacement part. Although these components are hard to service, they are all removable and therefore make them easy to replace or work around in the event that we must service another component around these.

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SUSPENSION, STEERING AND BRAKES

Requirements

Team Goals:

- Design a suspension system that integrates with steering or CV axles
- Design a suspension focusing on ride height, roll center, KPI, bump steer, camber and scrub radius

Research and Development

Research:

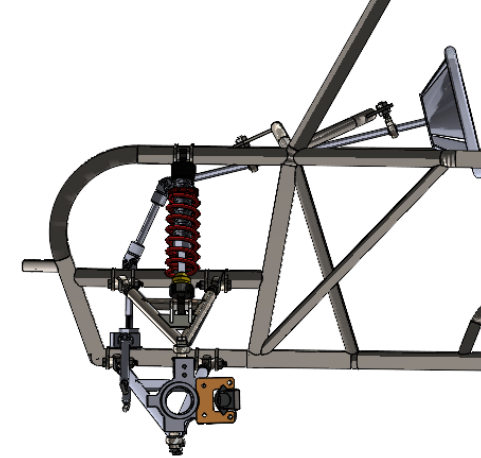
- The suspension arm design we are choosing to integrate is the double wishbone A-arm. This configuration offers better control over the kinematics than a strut-type arrangement.
- Ackerman steering geometry will be incorporated to steer the car.
- All brakes will be disc brakes and be locked outboard directly on the hubs
- 2 Separate master cylinders controlled by a portioning valve

SUSPENSION, STEERING AND BRAKES

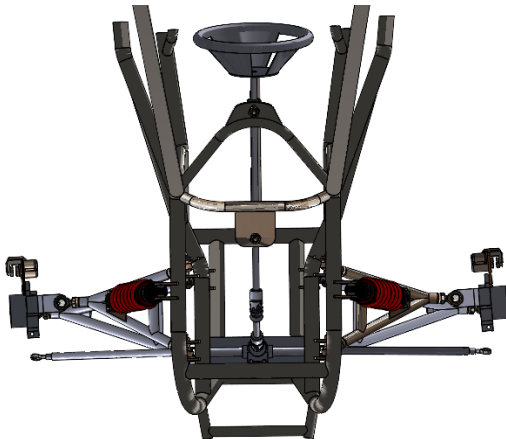
Front Suspension and Steering CAD – Front View



Front Suspension and Steering CAD – Left Side View



Front Suspension and Steering CAD – Top View

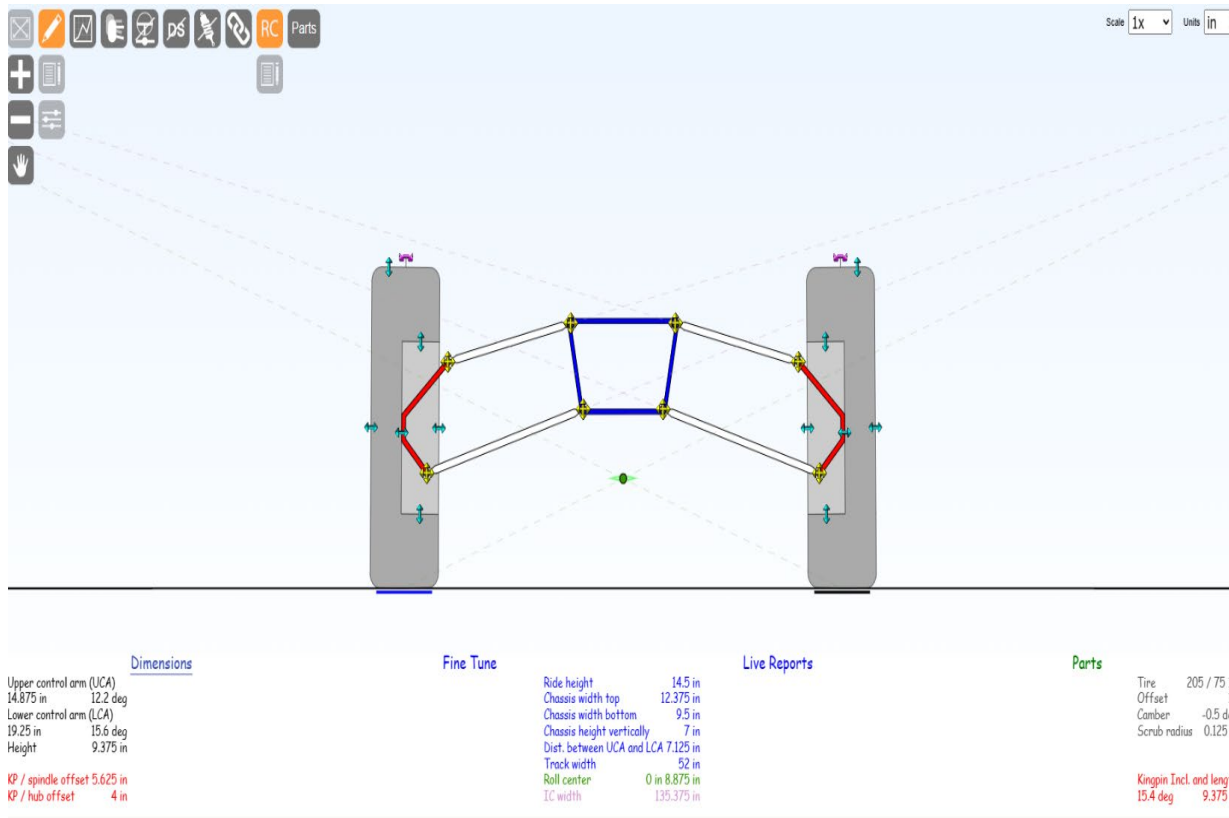


Front Suspension and Steering CAD – Isometric View



SUSPENSION, STEERING AND BRAKES

Front Suspension and Steering Data Collection for Analysis



Front Suspension and Steering Analysis Summary

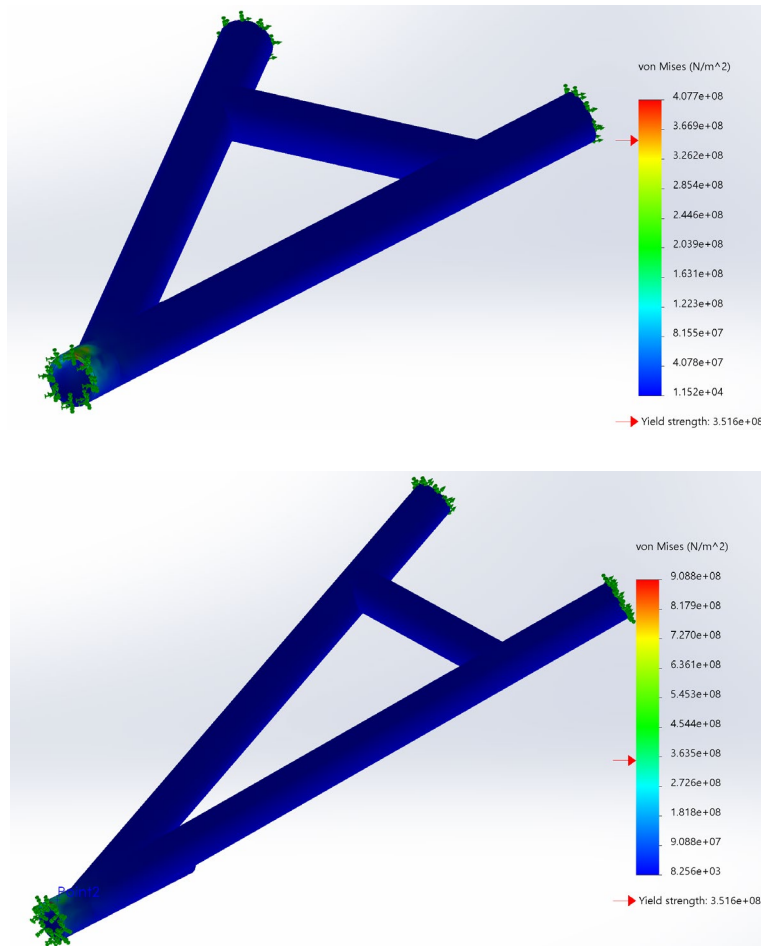
Analysis

- *Priorities while using aspirations racing suspension software:*
 - *Camber*
 - *Roll Center*
 - *Scrub Radius*
 - *KPI*
- *Roll Center was the target constraint to tie in all the priorities*

SUSPENSION, STEERING AND BRAKES

Front Suspension and Steering Testing / Validation

FEA was conducted on the front suspension

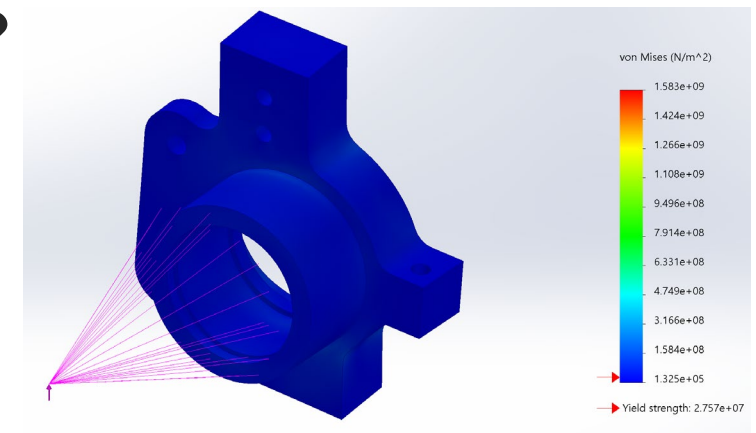


Front Suspension and Steering Correlation of Analysis and Testing

Are the tabs strong enough to withstand the forces imparted upon them?

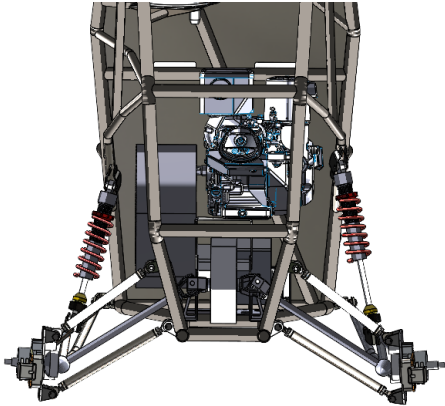
Does the suspension support the weight of the car?

Is the suspension soft enough to handle the track conditions expected?

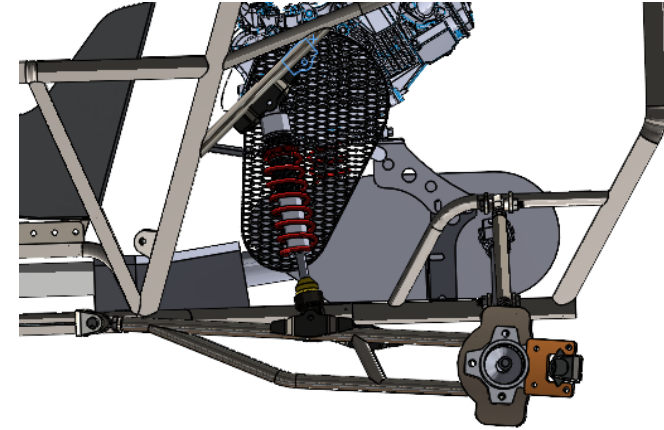


SUSPENSION, STEERING AND BRAKES

Rear Suspension CAD – Rear View



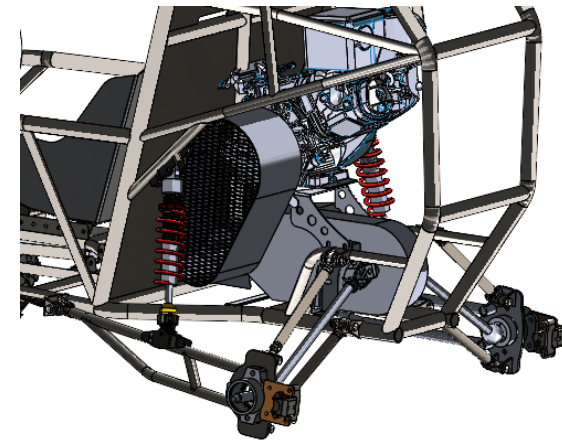
Rear Suspension CAD – Left Side View



Rear Suspension CAD – Top View

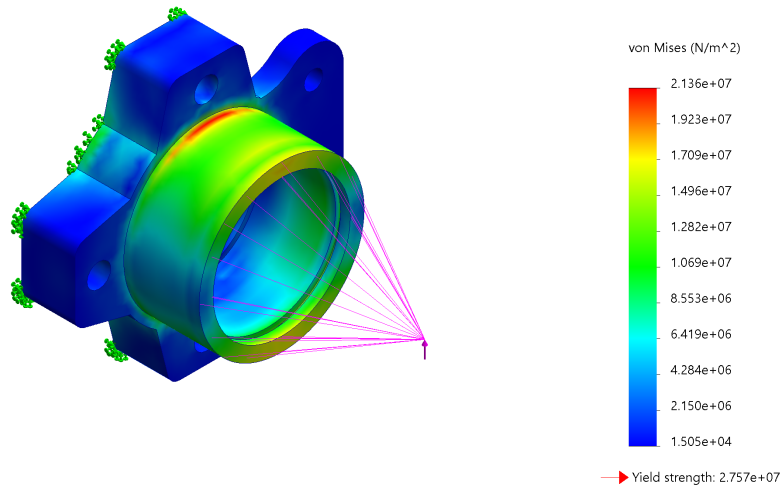


Rear Suspension CAD – Isometric View

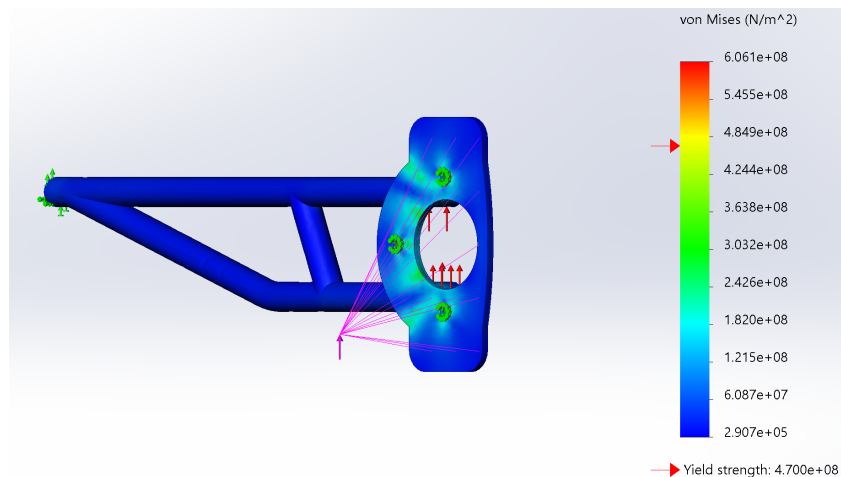


SUSPENSION, STEERING AND BRAKES

Rear Suspension Data Collection for Analysis



Rear Hub FEA Analysis



Rear Suspension Analysis Summary

Hub Carriers:

- A total of 2,500 lbf was applied where the force of rear bump loading would take place.

Trailing Arm:

- The analysis performed on the trailing arms included force on the plate where the arms attach and a remote load where the hub would be

SUSPENSION, STEERING AND BRAKES

Rear Suspension Testing / Validation

No testing has been performed on the rear suspension

Rear Suspension Correlation of Analysis and Testing

Testing to be done:

- *Does the axle maintain the same length throughout travel?*
- *Do the rear socks support the weight of the car?*
- *Does the tire maintain favorable conditions throughout travel?*
- *Are the rear wheels parallel at static ride height?*
- *Does the rear bearing carrier hold the car under dynamic loading?*

SUSPENSION, STEERING AND BRAKES

Brakes Testing / Validation

No testing currently has been conducted on the brakes.

Brakes Correlation of Analysis and Testing

Testing to be done:

- Test that all four wheels lock up
- Brakes must also work when the engine is off